



1

VIRTUALIZATION

Type-1 vs Type-2 Hypervisors

D1 — Virtualization



TYPE-1 — BARE-METAL

- Runs directly on physical hardware — no host OS underneath
- Attack surface = the hypervisor only. No OS layer to exploit.
- Better performance — no OS overhead
- Examples: VMware ESXi, Microsoft Hyper-V (Server), Xen

TYPE-2 — HOSTED

- Runs as an application on top of a host OS (Windows / Linux / macOS)
- Attack surface = hypervisor + entire host OS
- Easier to install — suitable for dev/test, not production
- Examples: VMware Workstation, Oracle VirtualBox, Parallels

EXAM SHORTCUT — Type-1 = better security — no host OS layer to compromise

Type-1 vs Type-2 Hypervisor

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Hypervisor Types — Type-1 (Bare Metal) vs Type-2 (Hosted)



EXAM TIP: Type-1 = hypervisor IS the OS. Type-2 = hypervisor is an app running on an OS. · Type-1 used in enterprise; Type-2 used by developers.

Memory Ballooning

D1 — Virtualization



WHAT TRIGGERS IT

- Host physical RAM is running low — hypervisor needs to reclaim memory
- Works without shutting down or pausing the guest VM

HOW IT WORKS (3 steps)

- 1. Hypervisor detects host memory pressure
- 2. Signals the balloon driver INSIDE the guest VM
- 3. Balloon driver inflates — claims pages inside the guest OS
- → Guest OS runs short on RAM → pages idle processes to disk

WHAT YOU SEE IN MONITORING

- Guest VM: HIGH disk I/O + LOW CPU utilisation
- This is expected behaviour — not a storage fault

EXAM SHORTCUT — Balloon driver lives **INSIDE** the guest. Symptom = guest writing to disk.

Practice Question 1

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QUESTION 1

A virtualization engineer notices that a guest VM running on a Type-2 hypervisor is generating heavy disk write activity despite minimal application workload. Monitoring confirms the hypervisor recently activated memory reclamation on that guest. Which component is directly responsible for the increased disk I/O?

- A. The virtual network adapter redirects memory overflow frames to the disk subsystem.
- B. The balloon driver inflates inside the guest VM, forcing the guest OS to page idle processes to disk.
- C. The hypervisor kernel directly swaps guest physical pages into the host swap partition.
- D. The guest BIOS reduces allocated vRAM and triggers an emergency checkpoint write.

Answer — Question 1

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ANSWER B

B. The balloon driver inflates inside the guest VM, forcing the guest OS to page idle processes to disk.

Why this is correct

The balloon driver is installed inside the guest OS. When the hypervisor signals it, it inflates by allocating memory within the guest — causing the guest OS to believe it is short on RAM and to page its own idle processes out to disk. High disk I/O with low CPU is the classic fingerprint of active ballooning.

Why the other options are wrong

- A — Network adapters have no role in memory management or disk paging.
- C — The hypervisor does not directly swap guest pages; that is the guest OS's job after ballooning signals it.
- D — BIOS cannot reduce vRAM at runtime; ballooning is a software driver mechanism, not a firmware action.

Practice Question 2

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QUESTION 2

A security team must choose a hypervisor architecture for a new multi-tenant production environment where VM isolation is a priority. One engineer recommends bare-metal deployment. Which statement best explains the security benefit of that recommendation?

- A. Bare-metal hypervisors include a built-in network firewall that blocks all inter-VM traffic by default.
- B. Hosted hypervisors share hardware directly with guest VMs, removing OS abstraction overhead.
- C. A bare-metal hypervisor eliminates the host OS layer, reducing the attack surface that a hosted hypervisor exposes.
- D. Hosted hypervisors require TPM hardware that bare-metal deployments bypass for faster VM startup.

Answer — Question 2

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ANSWER C

C. A bare-metal hypervisor eliminates the host OS layer, reducing the attack surface that a hosted hypervisor exposes.

Why this is correct

Type-1 (bare-metal) hypervisors run directly on physical hardware with no host operating system. This removes an entire software layer — and all its vulnerabilities — from the attack surface. In a hosted (Type-2) deployment, an attacker who compromises the host OS can potentially reach all guest VMs running above it.

Why the other options are wrong

- A — Bare-metal hypervisors do not include built-in inter-VM firewalls by default.
- B — It is the bare-metal hypervisor, not the hosted one, that eliminates OS abstraction overhead.
- D — TPM requirements are unrelated to the Type-1/Type-2 distinction and do not affect VM startup speed.

Storage Virtualization — Virtual Disks + Datastores

D1 — Virtualization



VIRTUAL DISK IMAGE FORMATS

- VMDK (VMware Virtual Machine Disk): virtual disk format used by ESXi / VMware Workstation
- VHD / VHDX (Virtual Hard Disk): Microsoft format used by Hyper-V and Azure
- Each virtual disk = a FILE on the host that contains the guest OS entire disk
- Guest OS sees a normal disk; hypervisor translates I/O to file operations on the host
- Enables: snapshots, live migration, cloning — because the disk IS just a file

EXAM SHORTCUT — VMDK = VMware. VHD/VHDX = Microsoft/Hyper-V. Both = disk-in-a-file managed by hypervisor.

DATASTORES — SHARED STORAGE FOR HYPERVISORS

- Datastore: shared storage pool accessible to one or more hypervisor hosts
- Stores virtual disk files (VMDK/VHD), ISOs, snapshots, templates
- Backed by: NFS/NAS (file-level), iSCSI/FC SAN (block-level), or local storage
- Shared datastore = multiple hosts can read/write → enables vMotion live migration
- Key point: the HYPERVISOR mounts and manages the datastore — not the guest OS

EXAM SHORTCUT — Datastore = where VMDKs live. Hypervisor mounts it. Guest OS never sees the physical storage.

Storage Virtualization — Virtual Disks + Datastores

D1 — Virtualization



WHY VIRTUAL DISKS DECOUPLE GUEST FROM HARDWARE

- Guest OS boots from VMDK/VHD as if it were a physical disk — hardware-independent
- Move the VMDK file = move the entire VM to any compatible hypervisor host
- Physical disk fails → storage redundancy at datastore level, not guest level
- Different from NAS (file shares to OS), DAS (direct physical attach), iSCSI (raw block to guest)
- Virtual disk format IS the virtualization layer for storage — hypervisor handles translation

EXAM SHORTCUT — VMDK/VHD decouples guest from physical disk. Portability + snapshot + migration all flow from this.

Practice Question 1.3

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QUESTION 1.3

Which storage model is specifically designed for virtualized environments to decouple guest OS storage from the underlying physical disk hardware, while enabling features such as live migration and instant snapshots?

- A. NAS using SMB/CIFS protocol sharing a traditional file system tree directly to each hypervisor node as a network drive.
- B. Virtual disk images (VMDK/VHD) stored in a shared datastore and mounted by the hypervisor on behalf of the guest OS.
- C. Direct-Attached Storage with RAID-5 striping across hot-swap spindle arrays connected directly to the server.
- D. iSCSI block storage accessed via a dedicated hardware SCSI controller installed inside the guest OS.

Answer — Question 1.3

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D1 — Virtualization

ANSWER B

B. Virtual disk images (VMDK/VHD) in a shared datastore — disk-in-a-file managed by the hypervisor.

Why this is correct

VMDK (VMware) and VHD/VHDX (Microsoft Hyper-V) encapsulate a guest's entire disk inside a file stored on a hypervisor-accessible datastore. The hypervisor mounts the datastore and presents the virtual disk to the guest, completely decoupling the guest OS from the physical hardware. This architecture is what makes snapshots, cloning, and live migration (vMotion) possible — the disk is just a file that can be copied, moved, or snapshotted atomically.

Why the other options are wrong

- A — NAS/SMB shares a filesystem to the guest OS as a network drive. This does not decouple the guest from hardware in the way virtual disk images do, and NAS shares are not the storage mechanism for running VM disk images.
- C — Direct-Attached Storage ties physical disks to a specific server with no abstraction layer. The opposite of decoupled — moving the VM requires physically moving or reconnecting the disks.
- D — iSCSI presents raw block storage. While iSCSI CAN back a datastore, iSCSI accessed directly inside the guest still ties the guest to the underlying storage technology and does not use the virtual disk image format (VMDK/VHD) that enables hypervisor-level features like snapshots and vMotion.